

Project Proposal Form, 2025
Institute of Information Technology (IIT)
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Project Title:	ResearchBridge: A Smart Research Collaboration Platform for Emerging Researchers		
Motivation: New researchers and graduate students face a critical three-part barrier: they cannot find relevant collaborators, lack shared tools to execute research projects, and have no guidance on the publication process. Existing platforms such as ResearchGate and Academia.edu are designed for researchers who already have credentials, not for those entering the pipeline. No single platform currently combines intelligent collaborator matching, a structured collaboration workspace, and a publication assistant in one system. This project proposes ResearchBridge, a microservices-based smart research collaboration platform that uses a Hybrid Collaborative Filtering + Content-Based (CF+CB) matching algorithm augmented by Sentence-BERT (Bidirectional Encoder Representations from Transformers) semantic embeddings to surface the right collaborators from day one. The platform solves the <i>cold-start problem</i> by bootstrapping from explicit profile attributes before transitioning to behavior-driven inference. A PageRank-variant TrustRank algorithm ensures community credibility, and a Publication Assistant module guides researchers from draft to submitted paper using journal databases (DOAJ/Scimago) and formatting templates.			
Project Description: 1. User Registration and Profile Management Users register with academic background, research interests, domain tags, skill declarations, and institution. Institutional email domains (<i>.edu</i>, <i>.ac.bd</i>) are verified automatically for trust tier assignment. Profiles are stored in PostgreSQL and indexed in Elasticsearch via dense vector			

embeddings for semantic search. This module is the foundation for all downstream matching and collaboration features.

2. AI-Powered Collaborator Matching Engine

The core algorithmic module. A **Hybrid CF+CB algorithm** runs on the **Python FastAPI ML microservice**. Content-Based Filtering (CBF) matches explicit profile tags for new users (solving the cold-start problem). Collaborative Filtering (CF) activates as behavioral signals (reads, bookmarks, comments) accumulate — the same model used by Netflix and Spotify. **Sentence-BERT** converts profile text and paper abstracts into dense vectors; cosine similarity search via **Elasticsearch kNN** and **Neo4j graph traversal** returns matched collaborators, relevant papers, and open projects in a unified recommendation feed cached in **Redis**.

3. Collaboration Workspace

Once a team is formed, the platform provides a dedicated workspace with a **task board**, **milestone tracker**, **real-time document co-editor**, and version history — all linked to the research project. Real-time synchronization is delivered via **Socket.IO** WebSocket connections. This eliminates the need for ad-hoc email threads, shared drives, and messaging apps by centralizing all project activity in one context-aware environment.

4. Knowledge Library

A shared repository where users upload and discover **papers, datasets, notes, and literature reviews**. All content is indexed in **Elasticsearch** for semantic full-text search. Papers are stored as PDFs in **Cloudflare R2 / Digital Ocean**. The library feeds the recommendation engine and grows richer with every upload, creating a compounding knowledge asset for the research community.

5. Community Forum with TrustRank Scoring

A Q&A and discussion forum where researchers share knowledge, conduct AMAs with professors, and review each other's work. Unlike simple upvote systems, the platform uses a **TrustRank algorithm (PageRank variant)** where credibility propagates through citation graphs and endorsements. Verified professor accounts carry higher authority, preventing spam and surfacing quality knowledge. Forum posts and answers are embedded in the same semantic space as profiles and papers, making them searchable alongside research content.

6. Mentorship Module

Professors and senior researchers can offer structured mentorship slots. The system pairs them

with junior researchers using the same CF+CB matching engine, filtered by domain, availability, and mentorship history. This addresses the **tacit knowledge gap** — the informal guidance on how research actually works that no documentation captures. Mentorship interactions earn reputation points visible on the researcher's profile.

7. Publication Assistant

Guides a researcher from draft paper to journal submission. The module queries **DOAJ** and **Scimago** databases to recommend suitable journals based on paper abstract, topic tags, and impact factor preferences. An **LLM layer** generates citation files in BibTeX/APA/IEEE format and provides writing feedback. A structured **pre-submission checklist** ensures the paper meets formatting, ethical, and structural requirements before submission.

8. Notification and Event System

Real-time and asynchronous notifications inform users of: new collaborator matches, workspace activity, forum replies, mentorship requests, and publication updates. Notifications are delivered via **Socket.IO** (live) and **Apache Kafka** event queues (async). The system ensures users take timely action without checking the platform constantly, improving engagement and collaboration momentum.

9. Researcher Profile and Credential Dashboard

Each researcher has a public profile that grows over time, displaying: papers uploaded and co-authored, collaborations formed, community contributions (answers, reviews), mentorship history, and publication achievements. Profile data is stored in **PostgreSQL** with an append-only audit log. This creates a **verifiable academic portfolio** that compounds in value with every platform interaction — analogous to a GitHub profile for research.

10. Admin Dashboard and Analytics

An administrative panel for platform managers and faculty coordinators to monitor system health: user activity, match quality metrics, collaboration success rates, publication outcomes, and community moderation. Weekly analytics reports are generated from aggregated data, providing insight into research trends, active domains, and platform growth. Admins can manage trust tiers, verify institutional badges, and moderate forum content.

Strategy & Timeline:

The development follows an **Agile, modular sprint-based approach**. Core infrastructure (authentication, profile, matching engine) is built first. Feature modules are added iteratively in dependency order. The architecture uses **Next.js** for the frontend (SSR for SEO), **Node.js + Express** for the API gateway, **Python FastAPI** as an isolated ML microservice hosting Sentence-BERT and the CF+CB algorithm, and a **Kafka**-driven async event pipeline for background processing. Data is stored across **PostgreSQL** (transactional), **Neo4j** (relationship graph), **Elasticsearch** (semantic search), and **Redis** (recommendation feed cache). The full stack runs locally via Docker Compose at zero cost during development.

The project Gantt chart is given below:

We ek	Task No / Module	Description	Dependencies
1	1. User Auth & Profile	Implement registration, login, JWT auth, role-based access	None
2	1. User Auth & Profile	Profile fields, institution verification, trust tier logic	None
3	2. Matching Engine — CBF	Content-Based Filtering: tag-overlap matching, cold-start handler	Task 1
4	2. Matching Engine — CF	Collaborative Filtering: behavior signals, Neo4j graph setup	Task 1
5	2. Sentence-BERT Integration	Embed profiles & papers; Elasticsearch kNN index; Redis feed cache	Task 1, 2
6	3. Collaboration Workspace	Task board, milestone tracker, document editor UI	Task 1
7	3. Real-time Sync	Socket.IO integration for live doc editing and notifications	Task 6
8	4. Knowledge Library	Paper/dataset upload, PDF storage (R2/S3), Elasticsearch indexing	Task 1
9	5. Community Forum	Q&A posting, threading, TrustRank scoring algorithm	Task 1, 2
10	5. TrustRank Refinement	PageRank-variant credibility propagation, professor badge verification	Task 9
11	6. Mentorship Module	Mentorship slot creation, CF+CB pairing, session tracking	Task 2, 9
12	7. Publication Assistant	DOAJ/Scimago journal query API, journal recommendation engine	Task 4, 8

13	7. LLM Citation & Feedback	LLM integration for citation generation (BibTeX/APA), checklist	Task 12
14	8. Notification System	Kafka async events, Socket.IO push notifications, email alerts	Task 3, 7, 11
15	9. Researcher Profile	Public profile dashboard, contribution history, credential display	Task 1, 5, 9
16	10. Admin Dashboard	Admin analytics panel, moderation tools, trust tier management	Task 1, 5, 9
17	10. Analytics & Reports	Weekly report generation, research trend analysis, usage metrics	Task 16
18	Integration & API Testing	End-to-end integration of all microservices, API contract testing	All tasks
19	Performance & Load Testing	Kubernetes autoscale testing, Redis cache validation, BERT latency	Task 18
20	Security & Auth Hardening	OAuth 2.0 SSO, JWT expiry, rate limiting, input sanitization	Task 18
21	Final Integration & UAT	User acceptance testing with classmates/professors, bug fixes	All tasks
22	Project Submission & Docs	Final report, presentation slides, README, API documentation	All tasks

Languages or Tools to be used:

Frontend: Next.js (React), Tailwind CSS, Socket.IO

Backend API: Node.js + Express

ML Service: Python + FastAPI, Sentence-BERT (Hugging Face Transformers)

Databases: PostgreSQL, Neo4j, Elasticsearch, Redis

Message Queue: Apache Kafka

Infrastructure: Docker, Kubernetes

File Storage: Cloudflare R2 / Digital Ocean

IDE / Editor: Visual Studio Code

Supervisor's Name:	Dr. Kazi Muheymin-Us-Sakib
Signature of the supervisor: _____ Date:	
Before the Midterm Presentation 1: I confirm that the progress is satisfactory and I am forwarding it for the midterm presentation. Signature of the supervisor: _____ Date:	
Proposal Presentation Feedback (if any) : 	
Midterm 1 Presentation Feedback (if any) : 	
Midterm 2 Presentation Feedback (if any) : 	